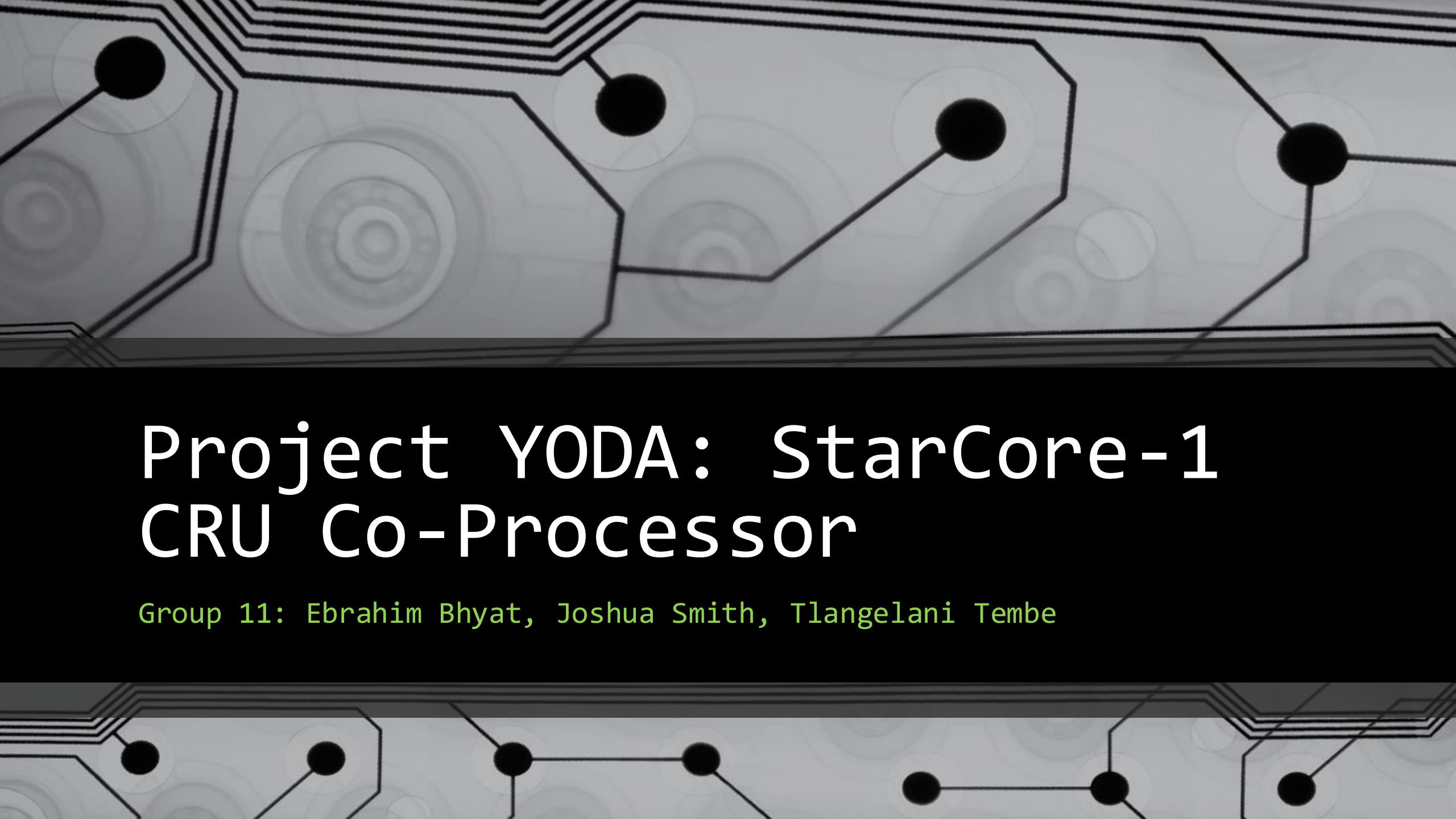


The background of the slide is a grayscale image of a circuit board. It features various traces, pads, and circular components. A solid black horizontal band runs across the middle of the image, serving as a backdrop for the title text.

Project YODA: StarCore-1 CRU Co-Processor

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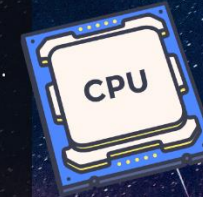
The CRU-16 Accelerator

StarCore CRU-16

“Hardware-Accelerated Trigonometry for the Stars”

Why Hardware Acceleration?

- Satellite altitude determination requires constant computation of sin and cos functions.
- Doing so in standard software using Taylor series approximations wastes clock cycles.
- Lookup tables on the other hand waste valuable storage.



The Solution: CRU-16

- The CRU-16 is a digital accelerator utilizing the Coordinate Rotation Digital Computer (CORDIC) algorithm to compute both Sine and Cosine simultaneously.
- The need for a hardware multiplier is completely bypassed as bit-shifts and additions are used for computation.

Technical Specifications

- **Precision:** Q2.14 Fixed-Point Architecture.
- **Speed:** Deterministic, executed in exactly 16 clock cycles.
- **Integration:** Interfaces seamlessly with StarCore-1.
- **Footprint:** Zero instruction memory overhead.



Limitations of the baseline StarCore-1

The goal: Continuous satellite altitude determination through direction Cosine matrices.

The obstacle: The baseline StarCore-1 lacks a hardware multiplier.

Software inefficiency: Computation of Sin and Cosine via Taylor series approximation requires many addition loops which takes many clock cycles per function.

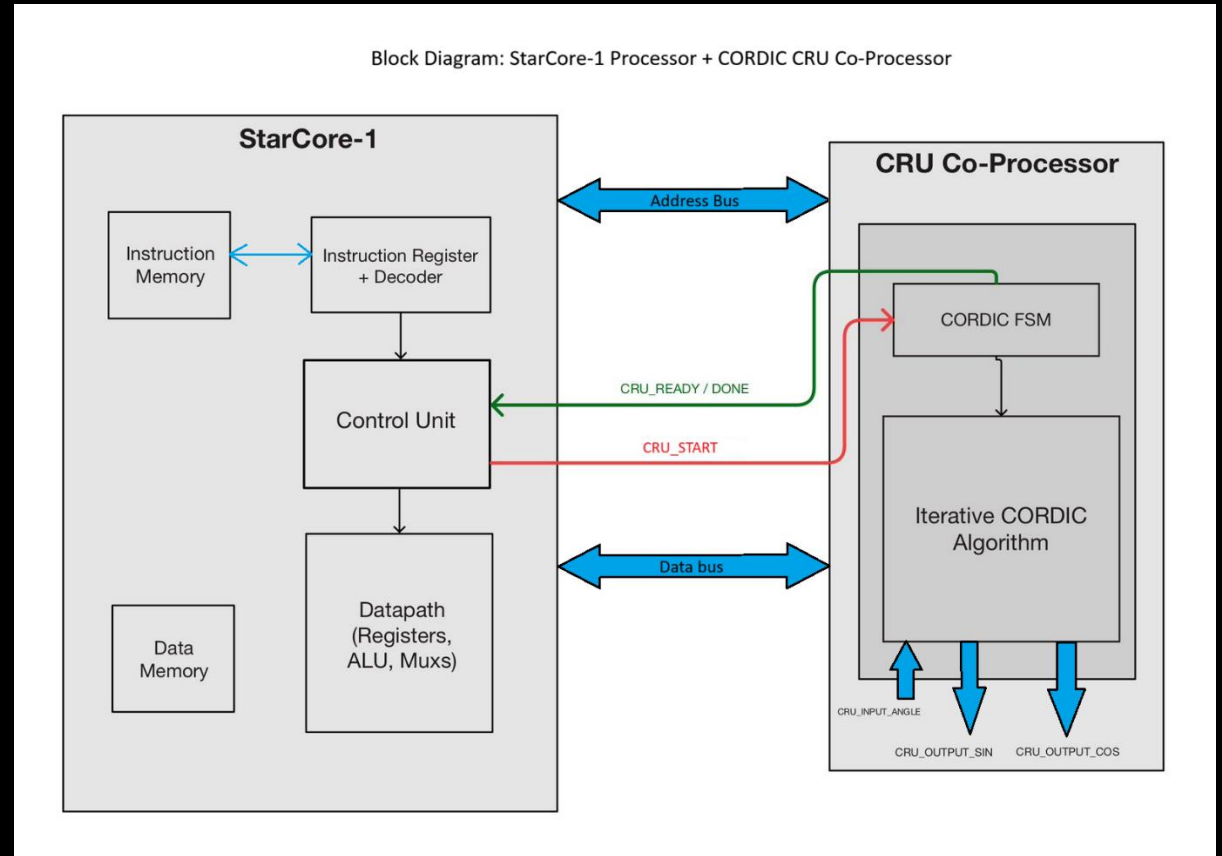
Memory inefficiency: Using pre-calculated look-up tables (LUTs) wastes instruction and data memory.

The Coordinate Rotation Digital Computer (CORDIC) Algorithm

- **Mode:** Rotation Mode (Calculates both Sin and Cosine)
- **Advantage:** Removes the need for a multiplier. Computation of trigonometry is done purely through bit shifts and addition.
- $X_{i+1} = X_i - d \cdot Y_i \cdot 2^{-i}$

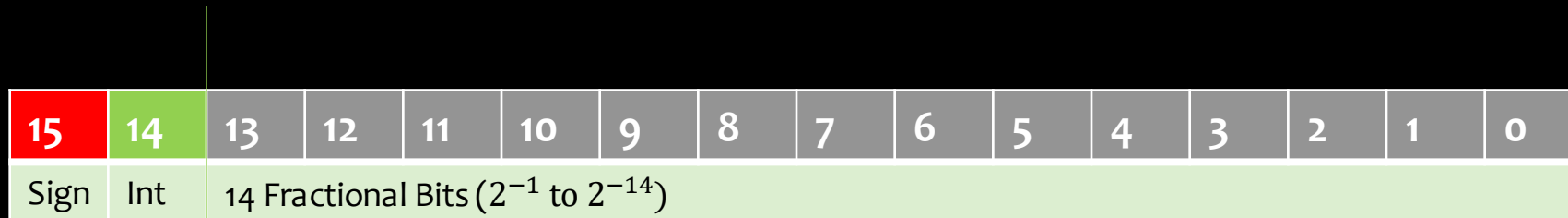
Core to Accelerator Integration

- Reserved opcode **1010** is decoded by the StarCore-1 Control Unit.
- **Interface:** Memory mapped I/O.
- **Handshaking:** Core writes the target angle, stalls execution, and waits for the CRU's done signal to read the results from the data bus.



Fixed-Point Internal Architecture

- **Constraint:** 16-bit register width.
- **Format:** Q2.14 Fixed-Point representation.
- **Breakdown:** 1 Sign bit, 1 Integer bit, 14 Fractional bits.
- **Resolution:** Capable of resolving trigonometric decimals to roughly 4 decimal places of accuracy.



CRU-16 Finite State Machine

Performance and Efficiency Gains

Verifying Q2.14 Hardware Constraints

- Limitations: 14 fractional bits gives a resolution of $1/16384$
- Problems
 - CORDIC needs acute angles in order to converge
- Solution
 - Input angle in 3.13 format
 - Apply trig identity in the form of reduction, and fold any angle into acute
- Result
 - Full 2π range
 - No change to core working

Golden Output Test Vectors

Hardware Simulation

Conclusion

- The CRU-16 successfully demonstrates the viability of the HPES Co-Processor path.
- Hardware-accelerated trigonometry provides critical performance gains for aerospace applications.
- System is fully verified mathematically and in RTL simulation.
- **Questions?**